

Jenna E. Moore

CONTACT INFORMATION	Duke University Department of Physics 120 Science Drive Durham, NC, 27708 USA	<i>Email:</i> jenna.moore@duke.edu <i>Web:</i> www.jenna-moore.com
RESEARCH INTERESTS	Astronomical instrumentation: developing and deploying cryogenic readout assemblies for highly multiplexed detector systems for next-generation cosmic microwave background (CMB) experiments; on-site systems integration. Microwave engineering: cryogenic testing and characterization of novel microwave/RF devices for CMB readout systems, including low-noise amplifiers. Cosmology and astrophysics: analysis of cosmic background radiation at microwave, optical, and infrared wavelengths	
EDUCATION	Arizona State University , Tempe, Arizona, USA Ph.D., Exploration Systems Design (Instrumentation) December 2023 Dissertation Topic: “Development and Integration of Readout Systems for the Simons Observatory and Measuring the Extragalactic Background Light from Optical Stacking of Quiescent Galaxies at Cosmic Noon” Dissertation Advisor: Philip Mauskopf Fisk University , Nashville, Tennessee, USA M.A., Physics July 2018 Thesis Topic: “Detection and Identification of Unknown Peaks within Gamma-Ray Spectra via Continuous Wavelet Transformation Analysis” Thesis Advisor: Arnold Burger Queens University of Charlotte , Charlotte, North Carolina, USA B.S., Chemistry <i>Magna cum laude</i> May 2016 Senior Capstone Research Topic: “A Simple LED Spectrophotometer for Demonstrating the Principles of Fluorometry and Colorimetry” Capstone Research Advisor: W. Andrew Tucker	
POSITIONS HELD	Postdoctoral Associate Oct 2024 – present Vavagiakis Lab Department of Physics Duke University, Durham, NC Postdoctoral Researcher Jan 2024 – Oct 2024 Astronomical Instrumentation Lab School of Earth and Space Exploration Arizona State University, Tempe, AZ Graduate Research Assistant Aug 2018 – Dec 2023 Astronomical Instrumentation Lab School of Earth and Space Exploration Arizona State University, Tempe, AZ	

Teaching Assistant

Fall 2019

SES 307: Spaceworks I
School of Earth and Space Exploration
Arizona State University, Tempe, AZ

Graduate Research Assistant

June 2016 – July 2018

Materials Science Applications Group
Department of Life & Physical Sciences
Fisk University, Nashville, TN

Chemistry Laboratory Assistant

Jan 2016 – May 2016

Department of Chemistry & Environmental Science
Queens University of Charlotte, Charlotte, NC

Supplemental Instruction Leader – Chemistry I and II

Aug 2015 – May 2016

Center for Student Success
Queens University of Charlotte, Charlotte, NC

**RESEARCH
COLLABORATIONS****Simons Observatory**

2018 – present

- Development and testing of cryogenic readout harnesses for small and large-aperture telescopes
- Testing of cryogenic low-noise amplifiers for Simons Observatory readout systems
- On-site integration at the high-elevation facility on Cerro Toco in Chile
- Remote observation coordinator

CCAT

2024 – present

- Review of cryogenic readout harness design for Prime-Cam
- Planning the on-site integration of Prime-Cam at the high-elevation facility on Cerro Chajnantor in Chile

PUBLICATIONS

- **Moore, Jenna E.**, Cohen, S., Mauskopf, P., and Scannapieco, E.
Integrated Extragalactic Light from Stacking 10^5 Random Pointings in the Dark Energy Survey Data,
submitted to ApJ, (2024).
- Sierra, C. E., and 44 other authors including **Moore, Jenna E.**
The Simons Observatory: Pre-deployment Performance of a Large Aperture Telescope Optics Tube in the 90 and 150 GHz Spectral Bands,
submitted to ApJS, (2024). arXiv: 2405.06868
- Galitzki, N. and 76 other authors including **Moore, Jenna E.**
The Simons Observatory: Design, integration, and testing of the small aperture telescopes,
ApJS 274(33), (2024). DOI: 10.3847/1538-4365/ad64c9
- Mangu, A. and 15 other authors including **Moore, Jenna E.**
The Simons Observatory: Design, Integration, and Current Status of Small Aperture Telescopes,
Proceedings of Science TAUP2023, (2024). DOI: 10.22323/1.441.0003
- Meinke, J., Cohen, S., **Moore, Jenna E.**, Böckmann, K., Mauskopf, P., and Scannapieco, E.
Evidence of Extended Dust and Feedback around $z \approx 1$ Quiescent Galaxies via Millimeter Observations,
ApJ 954(2), (2023). DOI: 10.3847/1538-4357/acdcf4

- **Moore, Jenna E.** et al.
Development and performance of Universal Readout Harnesses for the Simons Observatory, Proc. SPIE 12190. (2022). DOI: 10.1117/12.2630527
- Bhandarkar, T. and 18 other authors including **Moore, Jenna E.**
The Simons Observatory: Development and Validation of the Large Aperture Telescope Receiver, Proc. SPIE 12190. (2022). DOI: 10.1117/12.2629557
- Vavagiakis, E. and 35 other authors including **Moore, Jenna E.**
CCAT-prime: Design of the Mod-Cam receiver and 280 GHz MKID instrument module, Proc. SPIE 12190. (2022). DOI: 10.1117/12.2630115
- Xu, Z. and 44 other authors including **Moore, Jenna E.**
The Simons Observatory: the Large Aperture Telescope Receiver (LATR) integration and validation results, Proc. SPIE 11453. (2020). DOI: 10.117/12.2576151
- Zhu, N. and 70 other authors including **Moore, Jenna E.**
The Simons Observatory: the Large Aperture Telescope Receiver, (2020). arXiv: 2103.02747
- Rao, M. S., Silva-Feaver, M., and 34 other authors including **Moore, Jenna E.**
Simons Observatory Microwave SQUID Multiplexing Readout - Cryogenic RF Amplifier and Coax Chain Design, Journal of Low Temperature Physics 119, (2020). DOI: 10.1007/s10909-020-02429-y
- The Simons Observatory Collaboration (283 authors including **Moore, Jenna E.**)
The Simons Observatory: Astro2020 Decadal Project Whitepaper, white paper submitted to the Astro2020 Decadal Survey (2019). arXiv:1907.08284

PRESENTATIONS & POSTERS	Simons Observatory Collaboration Meeting, Chicago, IL	2024
	SPIE Astronomical Telescopes & Instrumentation XI, Montreal, CA	2022
	Simons Observatory Collaboration Meeting, San Diego, CA	2022
	SESE Research Symposium, Arizona State University, Tempe, AZ (virtual)	2021
	Fisk-Vanderbilt Bridge Research Symposium, Fisk University, Nashville, TN (invited alumni speaker)	2020
	Fisk-Vanderbilt Bridge Research Symposium, Fisk University, Nashville, TN	2017
	Science Awards Banquet, Queens University of Charlotte, Charlotte, NC (invited alumni speaker)	2017
	Marking Excellence Symposium, Queens University of Charlotte, Charlotte, NC	2016
SCHOLARSHIPS, FELLOWSHIPS, & COMPETITIVE AWARDS	Fisk-Vanderbilt Master's-to-PhD Bridge Fellowship	2016–2018
	Outstanding Student Award, Chemistry/Biochemistry, Queens University of Charlotte	2016
	Sara Doll Burgess Scholarship, Queens University of Charlotte	2013–2016
SERVICE & PUBLIC OUTREACH	Member, Talks Panel, Simons Observatory	2025-present
	Member, Organizing Committee, Duke University Cosmology Seminar	2024-present
	Member, Organizing Committee, Simons Observatory collaboration meeting (virtual)	2021
	Facilitator, Professional Skills Seminar, Fisk-Vanderbilt Bridge Program	2017-2018
	Intern, Project Scientist	2015
MENTEES	Qifeng Cheng, Duke Physics Ph.D. student	2024-present
	Yi Wang, Duke Physics Ph.D. Student	2024-present

Lalit Julapalli, NC School of Science and Mathematics student	2024-present
John Thomas, Vanderbilt University Physics REU student	Summer 2017
Richard Nederlander, Vanderbilt University Physics REU student	Summer 2017
Jasmin Tindal, Queens University of Charlotte undergraduate student	2016-2017
Kayley Marchena, Queens University of Charlotte undergraduate student	2015-2018

REFERENCES

Eve Vavagiakis – Postdoctoral Advisor

Assistant Professor, Department of Physics
Duke University, Durham, NC
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Philip Mauskopf – Ph.D. Advisor

Professor, School of Earth and Space Exploration and Department of Physics
Arizona State University, Tempe, AZ
Email: philip.mauskopf@asu.edu

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